Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam F · Form 2: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A student claims equal tension remains measurable after the ideal rope is cut between two freely hanging weights. Which change makes the original two-body solution invalid?

(A) Newton's third law stops applying

(B) The constraint tying the accelerations disappears

(C) Gravity changes for both masses

(D) The two masses become equal

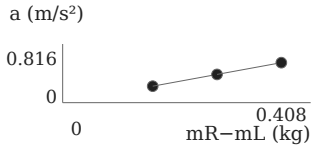
2. Transfer mass between two sides of an ideal Atwood pulley, keeping total mass 5 kg. Right downward is positive; g=10 m/s². Add equal masses to both pans to double total mass, testing the same imbalances. Which X in the modified apparatus yields Y=0.288?

(A) -0.552

(B) 1.548

(C) 0.288

(D) 0.708



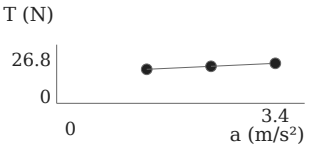
3. The left mass on an ideal pulley is 2 kg; vary the right mass. Left upward is positive; use g=10 m/s². a (m/s²) -> T (N): (1.4, 22.8); (2.4, 24.8); (3.4, 26.8). Both sensors multiply their axes by the same unknown positive gain k. What is the uncorrected slope?

(A) 1.2

(B) 2

(C) 3.6

(D) 2.8



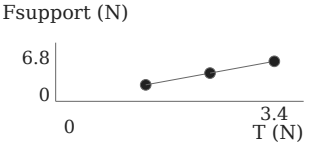
4. A negligible-weight fixed pulley has two vertical rope segments below it. A support holds its axle. Arrange the strands symmetrically at 60° from vertical instead of vertical. Plot the difference Ynew-Yold versus X. What slope should be observed?

(A) -0.58

(B) -1.84

(C) 0.26

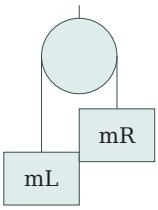
(D) -1



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Reverse mass imbalance Delta m at fixed total mass S in an Atwood machine. Its acceleration sensor has an unknown constant offset b. Keep right-down positive in both runs.

a. Use one positive direction for both runs and swap the heavier side in your sketch. **(1 pt)**



b. Derive both bias and g from the readings and identify the limit at Delta m=0. **(3 pt)**

6. A cart and a hanging mass share a taut rope over one ideal fixed pulley. Measure downward hanger displacement and the cart distance from the pulley. Delta yh (m) -> dcart (m): (1.4, 6.6); (2.4, 5.6); (3.4, 4.6).

a. Build the model for rope constraint from physics and predict both slope and intercept. **(2 pt)**

b. Double the initial cart distance at the same hanger displacements. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. During a pulley experiment a cart slides right against a constant resistance f. Students mistakenly add hanger mass m without removing any mass from the cart of fixed mass M. Derive a linear plot using measured m and a that can reveal both M and f. **(3 pt)**